

Quoc Anh Trinh (Andrew)

Data Analyst

M.Sc in Business Analytics, Deakin University

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Professional Summary

Data Analyst with 5+ years' experience in FinTech and banking, specialising in SQL- and Python-driven statistical analysis, transaction monitoring, and Power BI business intelligence dashboarding for leading institutions including Bendigo & Adelaide Bank and Remitano. Applies advanced statistical modelling, machine learning, NLP, and Generative AI / LLM solutions on AWS and GCP, building ETL pipelines and data-driven tools that reduced false positives by up to 67%, cut regulatory reporting time by 80%, and lifted marketing conversion by 30%. Passionate about translating complex data into measurable business outcomes across compliance, cost, and performance.

Technical Skills

- **Languages & Querying:** SQL, Python, R
- **Analytics & Machine Learning:** Machine Learning, NLP, Predictive Modelling, Statistical Calibration, Optimisation Algorithms, Scikit-learn, PySpark, Generative AI, LLMs, Agentic AI
- **BI & Visualisation:** Power BI, Tableau, Metabase, Excel
- **Data Engineering & ETL:** dbt, BigQuery, Amazon Redshift, AWS Glue, Apache Spark, Vector Databases
- **Cloud & Platforms:** Amazon Web Services (AWS), Google Cloud Platform (GCP), FastAPI, Next.js
- **Domain Expertise:** AML/CTF, Transaction Monitoring, Financial Crime Analytics, AUSTRAC Regulatory Reporting, Suspicious Matter Reports (SMR)

Professional Experience

Analyst, Intelligence and Analytics - Financial Crime Risk – Bendigo & Adelaide Bank 05/2021 - Present

Australian banking group with 4 commercial banks and 1 digital bank under management

- Own the transaction monitoring analytics function, serving operations, transformation, compliance, assurance, and governance teams.
- Own the Power BI dashboards within the transaction monitoring product, designing, developing, and maintaining them to provide key measures, KPIs, and guardrails.
- Research and design new transaction monitoring detection scenarios, then build and fine-tune them using statistical and machine learning techniques.
- Built vector search infrastructure and a real-time streaming UI to power an AUSTRAC Knowledge Hub and AML/CTF Program Review Hub, integrating enforcement action statements to automatically detect and evaluate regulatory gaps.
- Led the establishment of an interim fraud and AML transaction monitoring function during the bank's exit from DELPHI and transition from TechOne, engineering advanced SQL-based detection queries to maintain 100% monitoring continuity across 2M+ transactions with zero compliance breaches.
- Co-designed and delivered an end-to-end AI-driven UAR-to-SMR pipeline implementing secure data masking/rehydration and Pydantic-validated LLM prompting, reducing regulatory reporting preparation time by 80% while ensuring strict transparency and data privacy compliance.
- Developed and fine-tuned new transaction monitoring scenarios for Child Exploitation, Sextortion, Drug Trafficking, and Terrorism Financing typologies, leveraging genetic algorithms, statistical calibration, and trend modelling to achieve 60-70% true positive rates with a 67% reduction in alert volume.

- Researched and designed 9 new detection scenarios, applying statistical analysis to calibrate each to an effective 70% false positive baseline, maximising detection coverage while avoiding overly narrow scope.
- Engineered a typology extractor using Agentic AI and advanced ETL to transform 156 AUSTRAC guidance documents into structured data for transaction monitoring framework comparison and national monitoring.
- Built complex SQL and Python-based transaction monitoring scenarios outside vendor platforms and engineered automated daily workflows, processing 150+ alerts daily with 99.9% data integrity and eliminating manual transfers.
- Led the annual transaction monitoring performance review, applying statistical confidence testing, data sampling, and SQL-based scenario recalibration to reduce false positives by 5% and remove 5,000+ unnecessary alerts annually, saving 300+ investigation hours per year.
- Engineered SQL-based replication and assurance testing of 40+ transaction monitoring scenarios for the Adelaide Bank to Bendigo Bank migration, achieving 100% scenario alignment and seamless model transfer.
- Designed and developed an AML analytics workbench on Google Cloud Platform using dbt and Big-Query to support external transaction monitoring model validation, with modular, version-controlled data transformation layers for scenario testing, threshold calibration, and model performance benchmarking.

Data Analyst – Remitano

03/2020 - 04/2021

International cryptocurrency exchange that matched \$500M+ in trading value across 30+ countries

- Handled all data analytics requests, delivering dashboards and diagnostic reports using Python, SQL, and Metabase.
- Designed, developed, and managed data crawlers, ETL pipelines, and Redshift database structures, and collaborated with engineering teams on complex SQL-based data transformations.
- Automated email marketing via customer classification models (Python/SQL), boosting click-through rates by 30% and enabling scalable campaign personalisation.
- Optimised ETL pipelines, indexing, and data normalisation in Redshift, reducing developer workload by 25%, cloud costs by 30%, and dashboard runtime by 70%.
- Delivered 100% on-time analytical reports and dashboards with consistent stakeholder satisfaction scores of 8/10 or higher.
- Engineered complex SQL solutions (including recursive queries) for MLM commission calculations, processing 150K+ referral relationships across 4 levels, eliminating the need for a graph database and powering a 4-tier referral program that drove 25% of new user growth.

Investment Associate – National Citizen Bank

09/2014 - 02/2018

Commercial bank with more than 20 years of operation

- Built financial models for property project estimation, company valuation, and debt restructure planning, and presented recommendations to the Investment Council.
- Coordinated with in-house executives and clients on lending due diligence and gathering business information for financial analysis and valuation.
- Secured Investment Council approval for 80% of recommendations, generating a public equity portfolio CAGR of 10%.
- Reduced investment approval time by 72 hours and eliminated major process conflicts by drafting the investment procedure with detailed accounting, authority-level, and document-revision controls.

Education

- Master of Business Analytics, Deakin University, 2019

- Bachelor of General Business Management, National Economics University, 2013

Certifications

- CFA Level 1 Passed, CFA Institute, 2013
- Google Cloud Platform Big Data and Machine Learning Fundamentals, Coursera, 2020
- Modernizing Data Lakes and Data Warehouses with GCP, Coursera, 2020
- Building Batch Data Pipelines on GCP, Coursera, 2020

Honours & Awards

- Third Place & Honourable Mention, Project SafeAus, Team VAIT, GovHack Australia, 2021
- #13 public / #32 private, PetFinder Competition, Team Vicohub-1, Kaggle, 2019
- First Prize & Runner-up, Programming Competition, Department of Education and Training, 2007-2009

Additional

- Tutored 100+ students in SQL and Python; Coordinator of the CFA Vietnam volunteer team.
- Won 2 Open Government hackathon prizes; build data-driven solutions for micro-businesses.